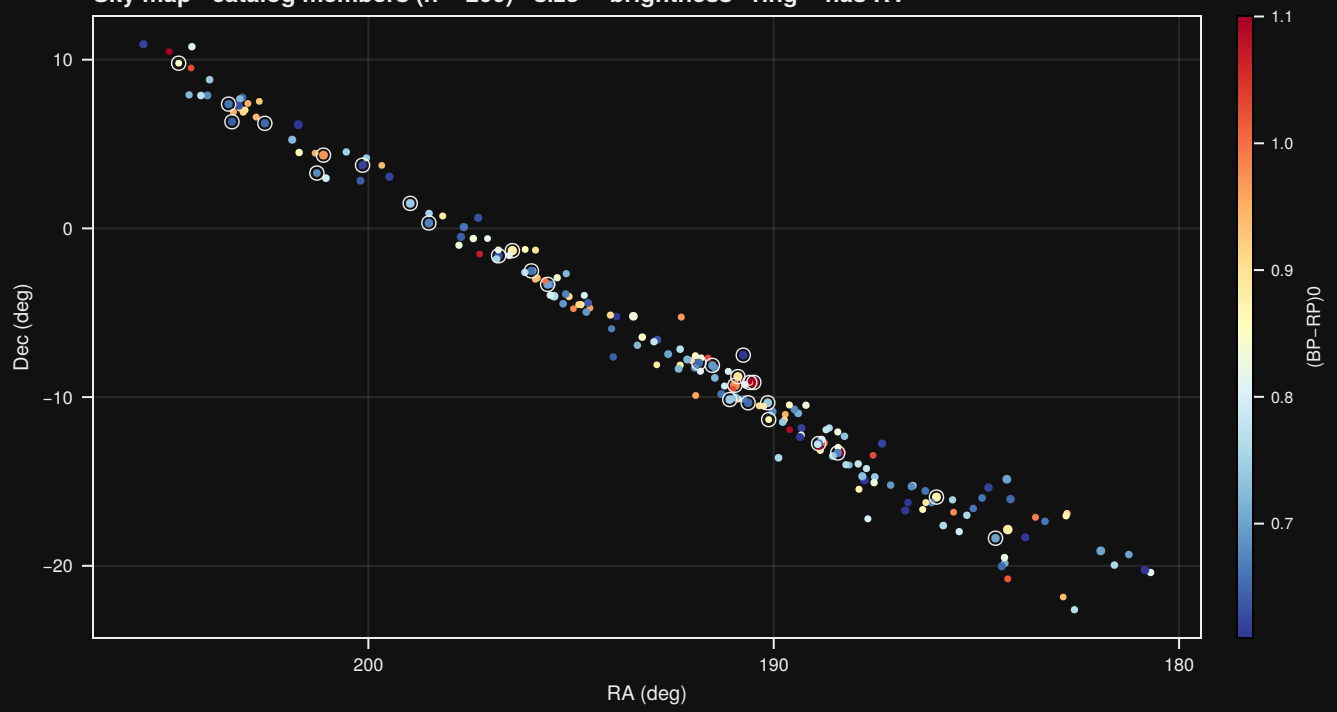
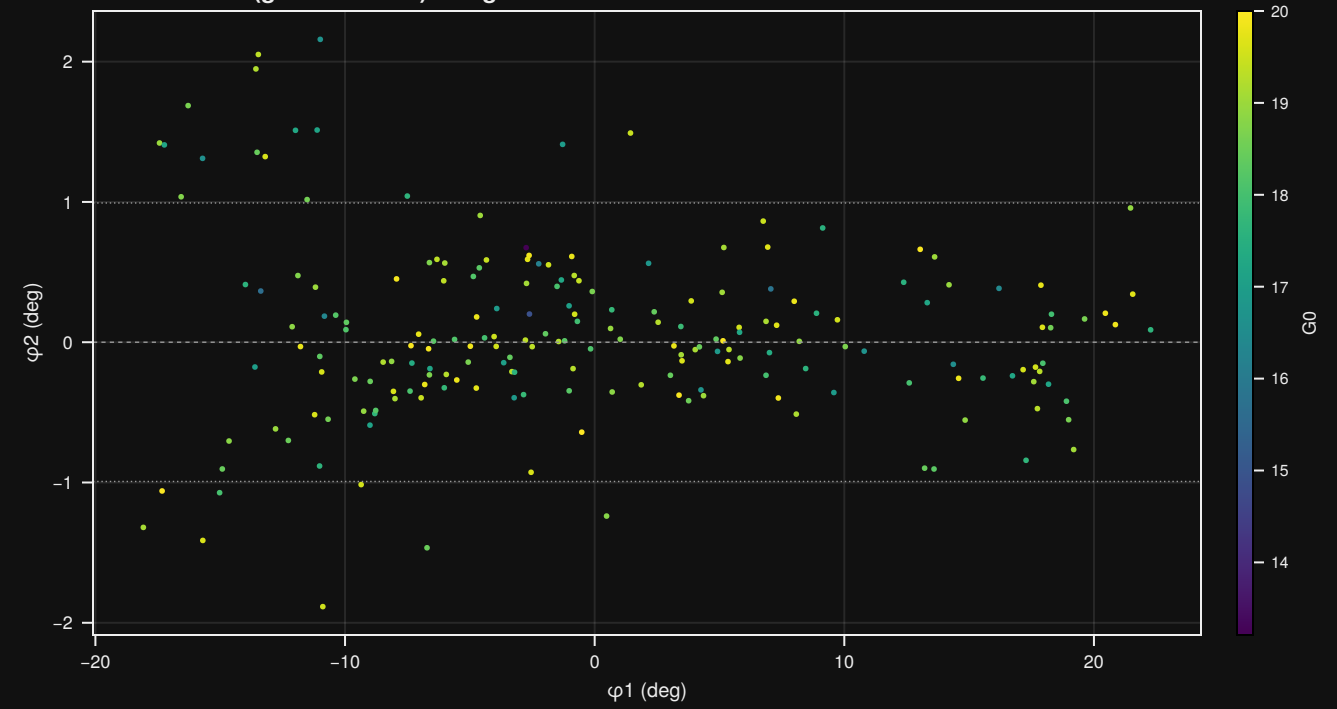
Gaia-1

MEMBERSHIP	
Stars in catalog	200
with Gaia GSP-Phot	108 (54%)
with radial velocity	28
Stream id (sID)	49
GEOMETRY · great-circle fit	
Centroid RA, Dec	192.5, -8.4 deg
Centroid l, b	305.0, 54.5 deg
Pole RA, Dec	107.8, 35.2 deg
Length (φ1, 2–98%)	37.0 deg
Width σ(φ2), robust	0.50 deg
KINEMATICS	
pmRA* med ± σ	-14.42 ± 1.55 mas/yr
pmDE med ± σ	-19.55 ± 1.72 mas/yr
μφ1* med ± σ	-24.42 ± 2.35 mas/yr
μφ2 med ± σ	0.35 ± 0.69 mas/yr
Parallax (cat.) med	0.186 ± 0.191 mas
Parallax (wtd, ZP-corr.)	0.197 ± 0.013 mas
Distance	d = 5.1 ± 0.3 kpc
Physical length	3.28 kpc
RV median ± σ(MAD)	195.6 ± 95.1 km/s
RV gradient dV/dφ1	-9.81 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.2–20.0 (18.8)
(BP−RP)0 median	0.77
GSP-Phot [M/H] (n=108)	-1.53 ± 0.44
GSP-Phot Teff median	5922 K
RUWE > 1.4	0.9 %
PM clusters (largest)	5 (94 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	200 | 200
n_RV catalog | paper	28 | 28
<[Fe/H]> paper	-1.80
Stellar mass (paper)	1100 Msun
Pole offset vs paper	0.3 deg

Sky map · catalog members (n = 200) · size ∝ brightness · ring = has RV

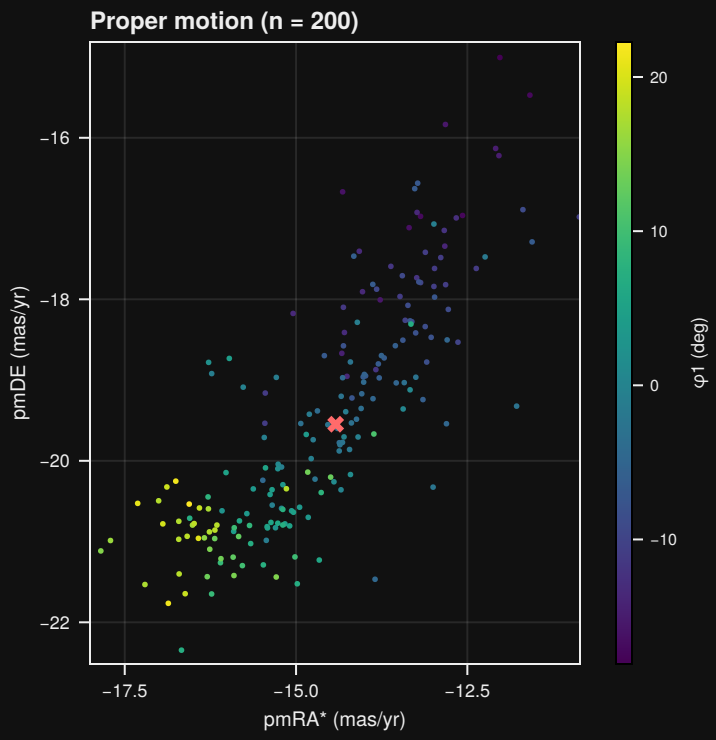


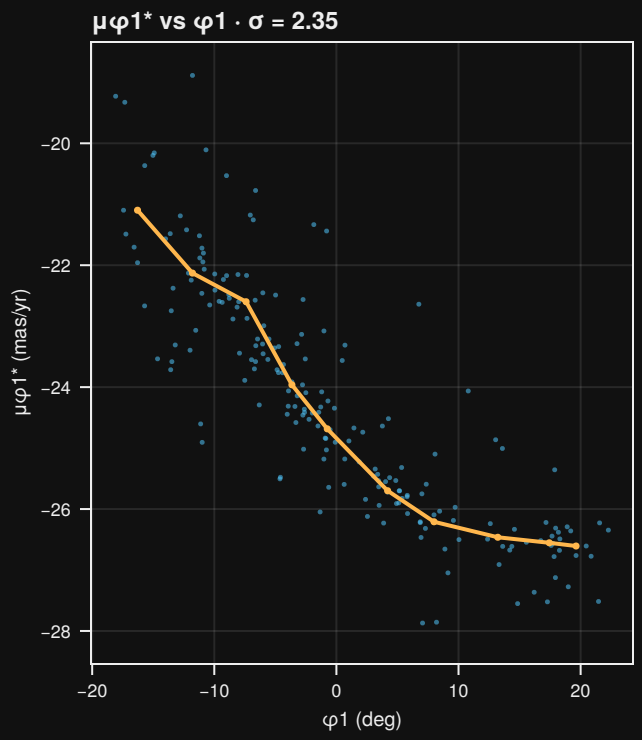
Stream frame (great-circle fit) · length 37.0° · width σ 0.50°

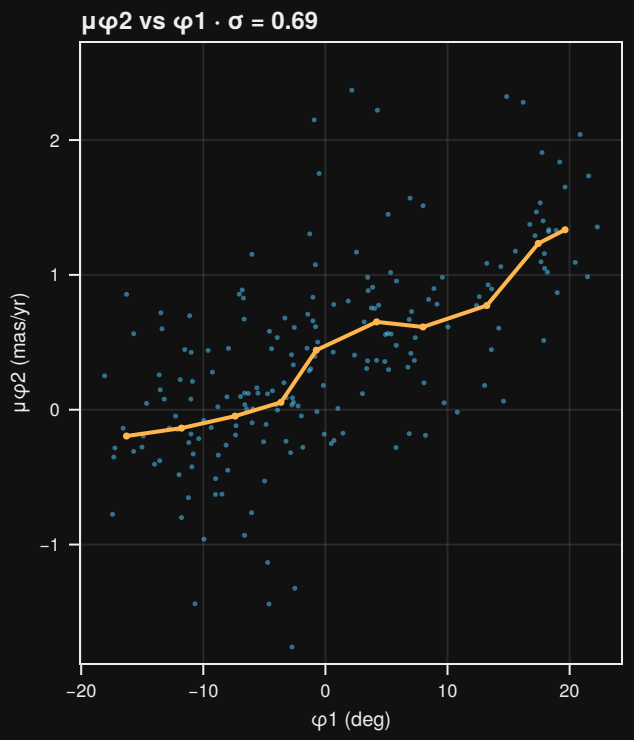


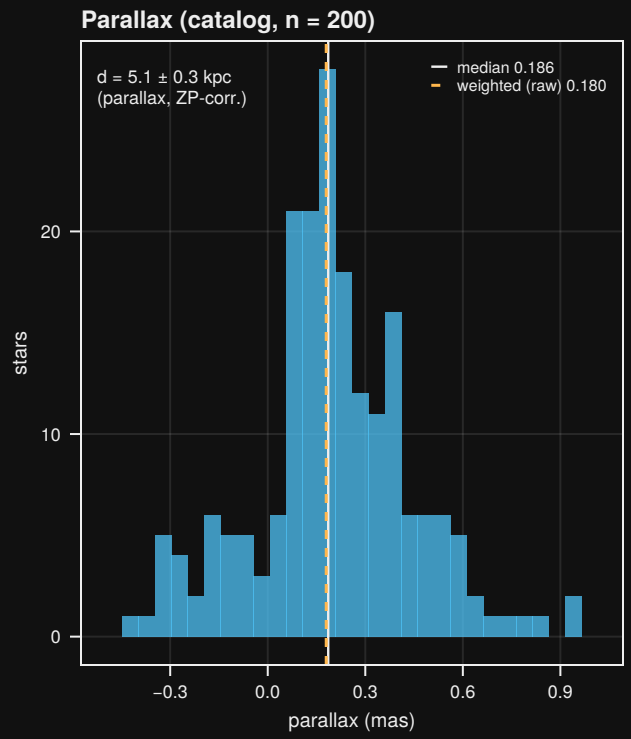
Gaia-1

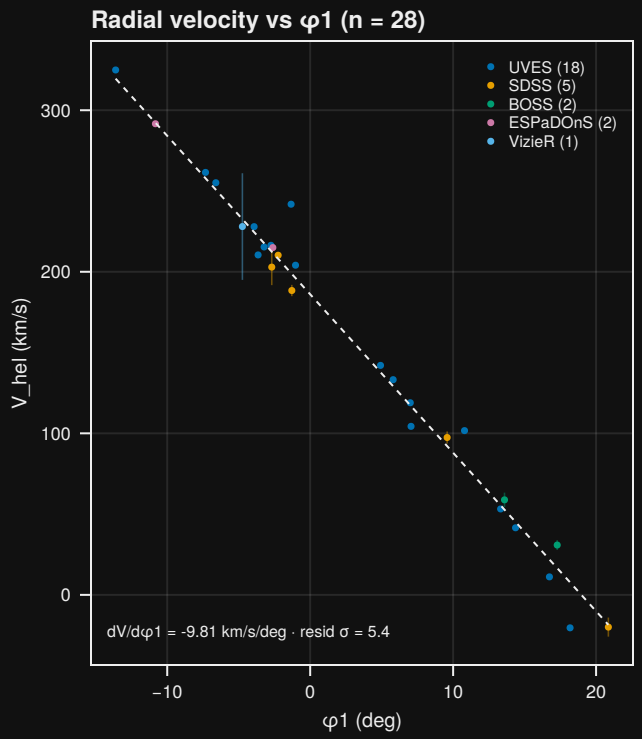
Kinematics · proper motion · parallax · radial velocity

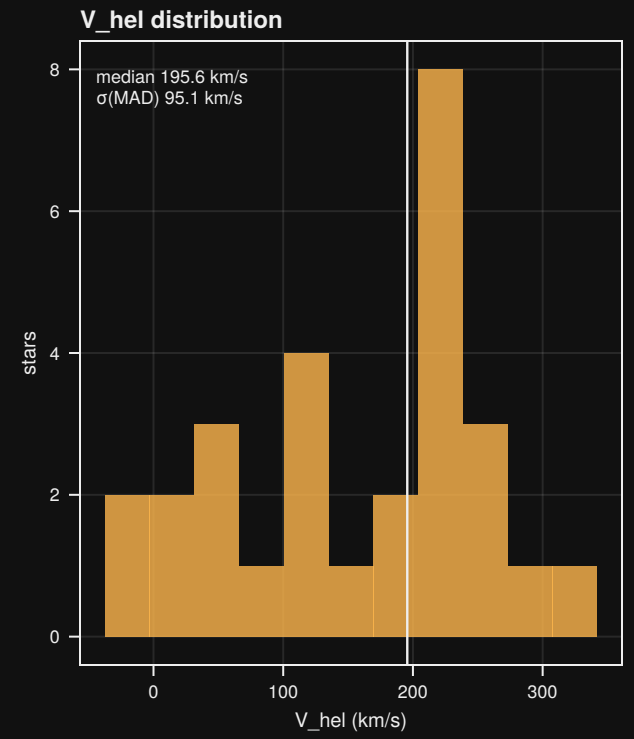






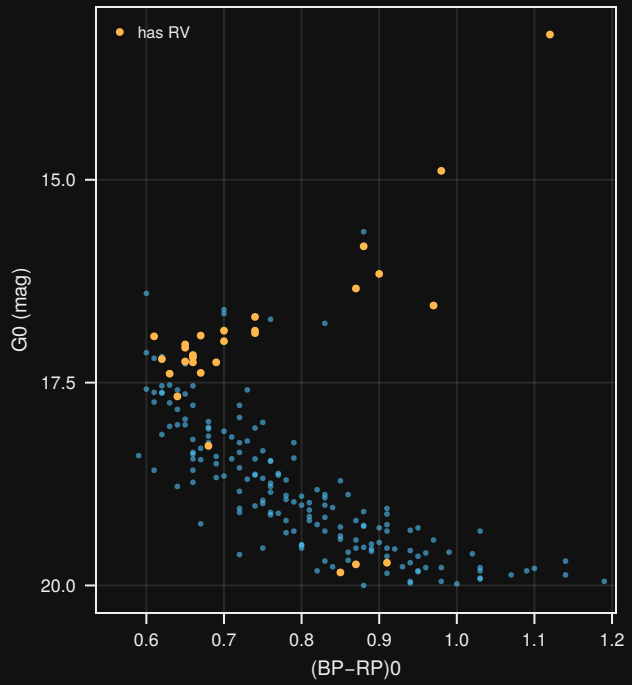




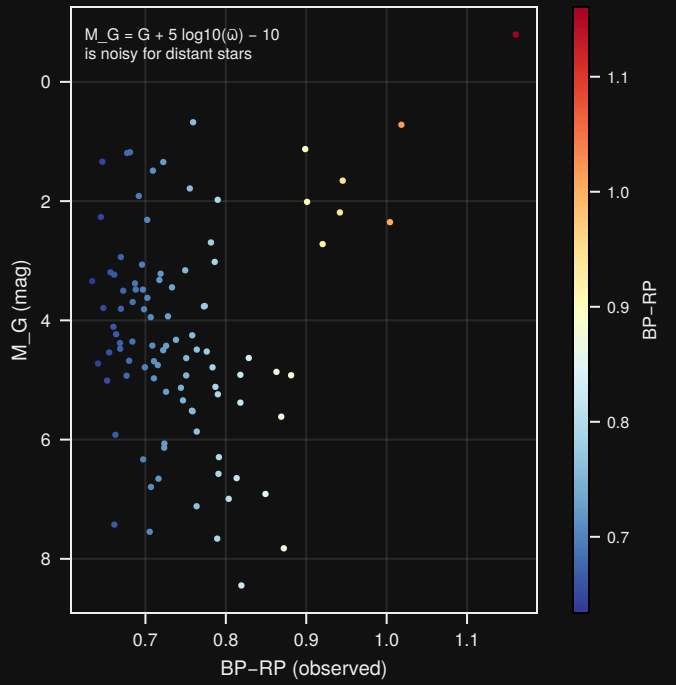


Gaia-1

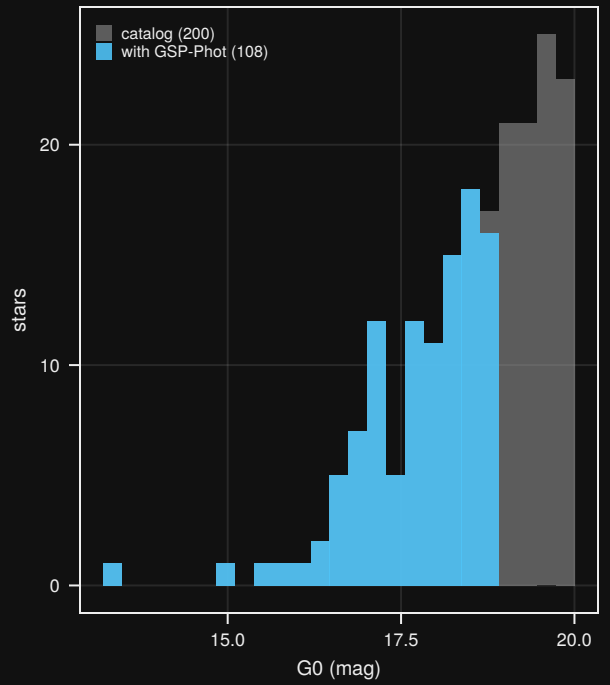
CMD · dereddened (n = 200)



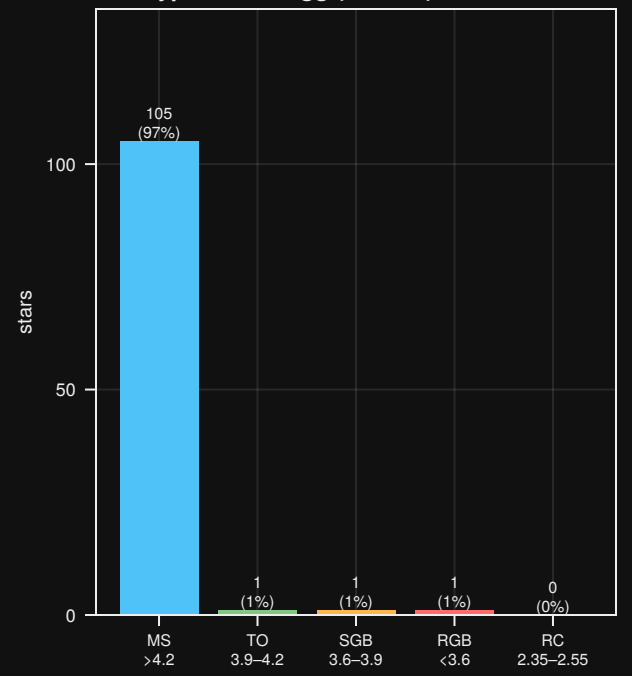
HR diagram · ϖ>0 (n = 98/108)



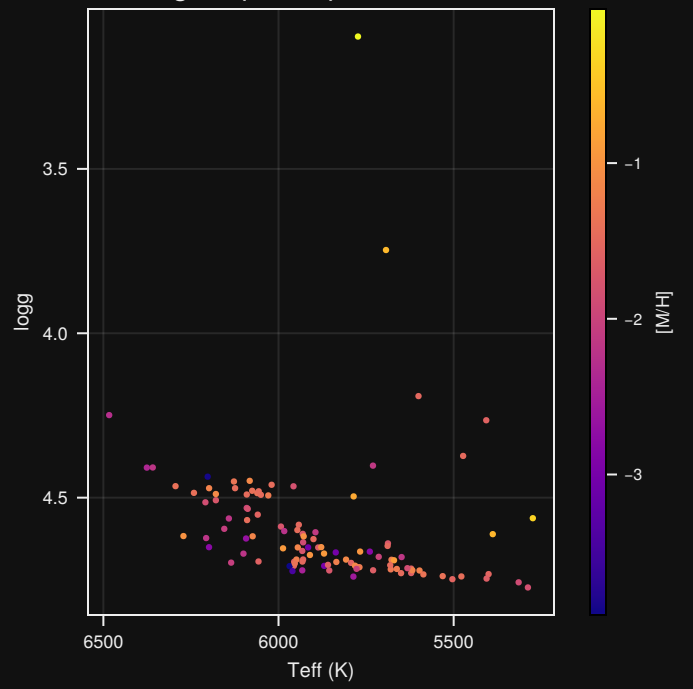
G0 & GSP-Phot coverage



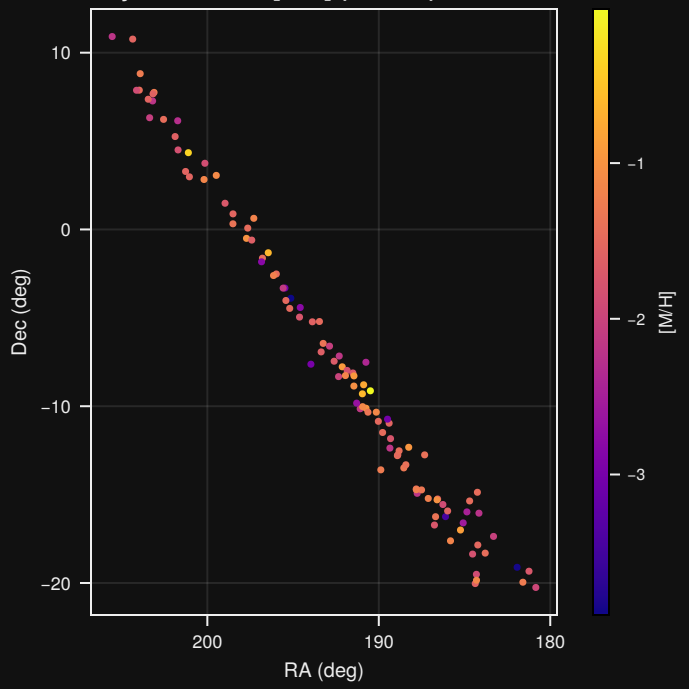
Star types from logg (n = 108)



Kiel diagram (n = 108)



Sky · GSP-Phot [M/H] (n = 108)

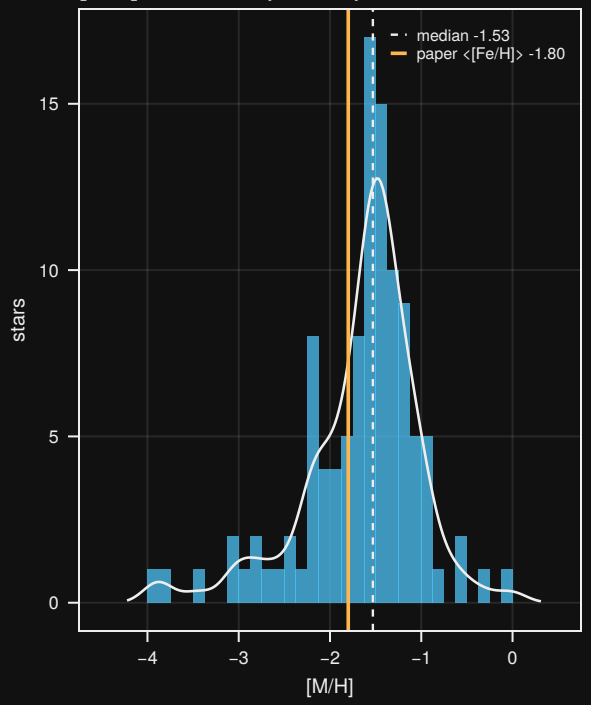


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

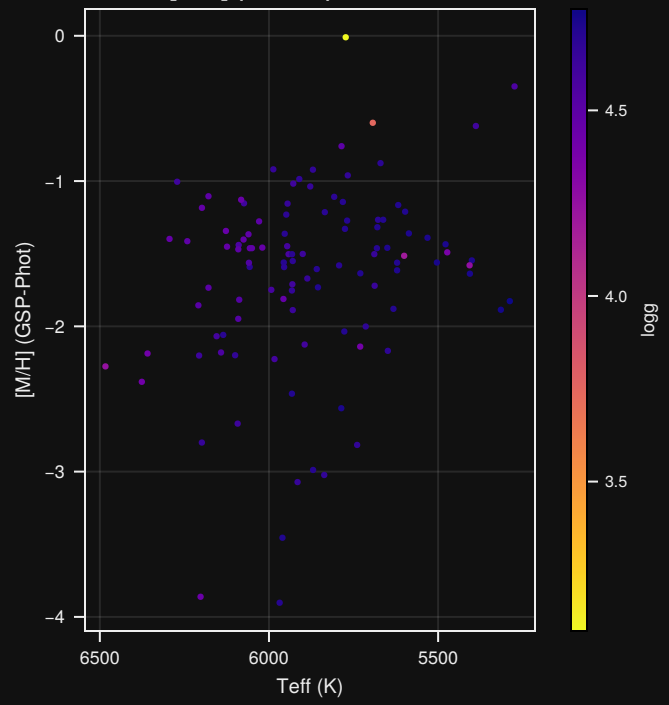
Gaia-1

Metallicity · age · astrometric quality

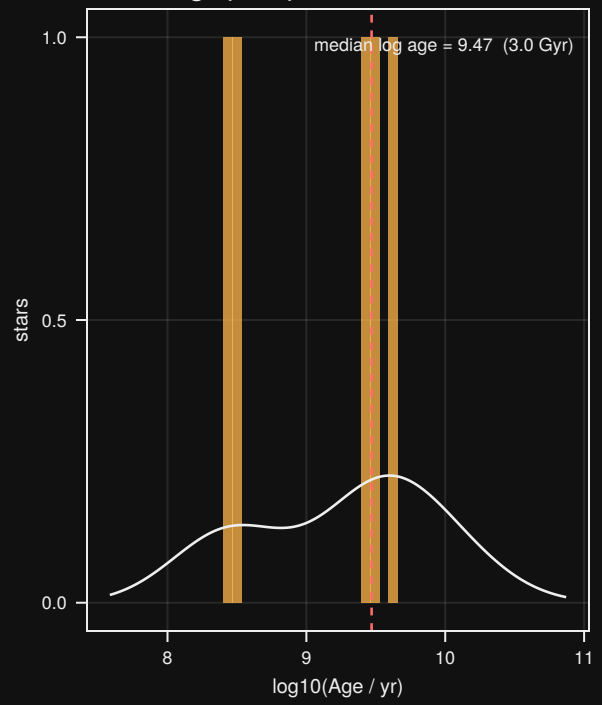
[M/H] GSP-Phot (n = 108)



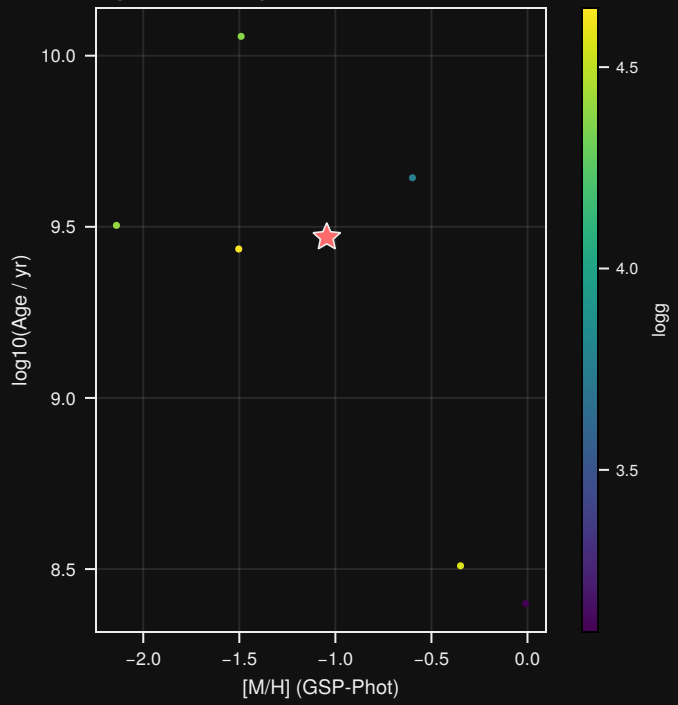
Teff vs [M/H] (n = 108)



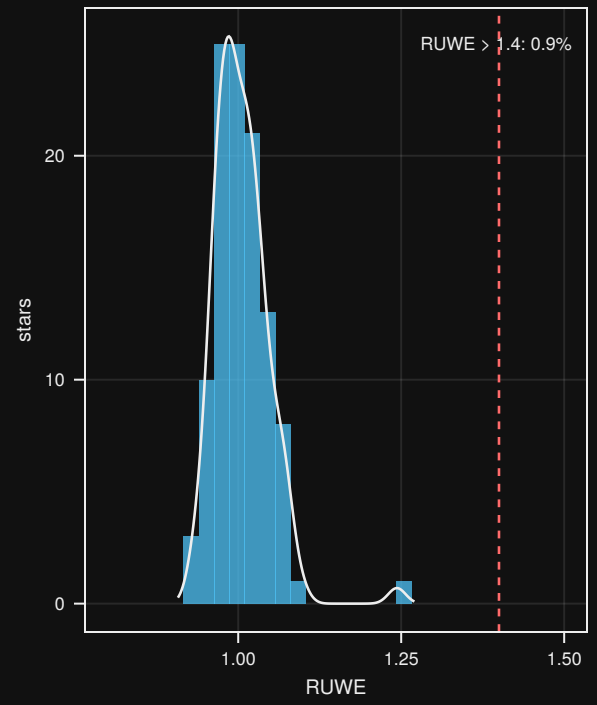
FLAME age (n = 6)



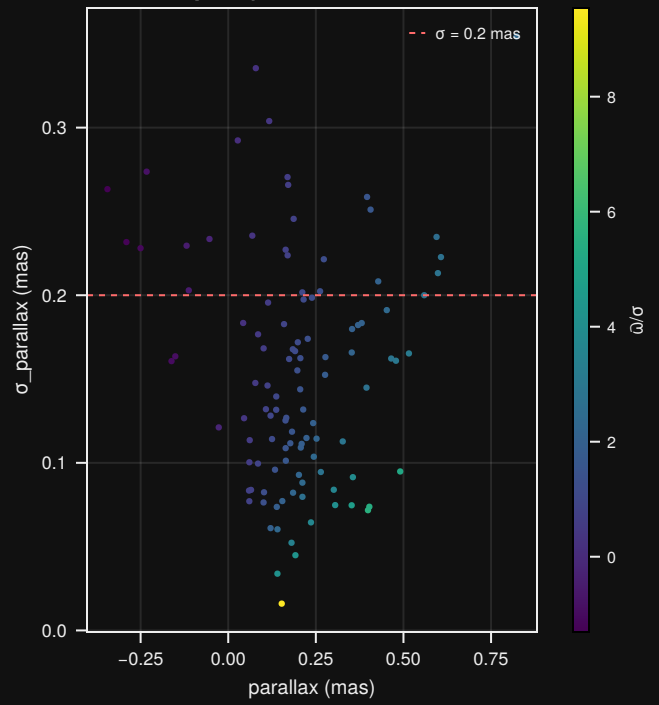
Age–metallicity (n = 6)



RUWE (n = 107)

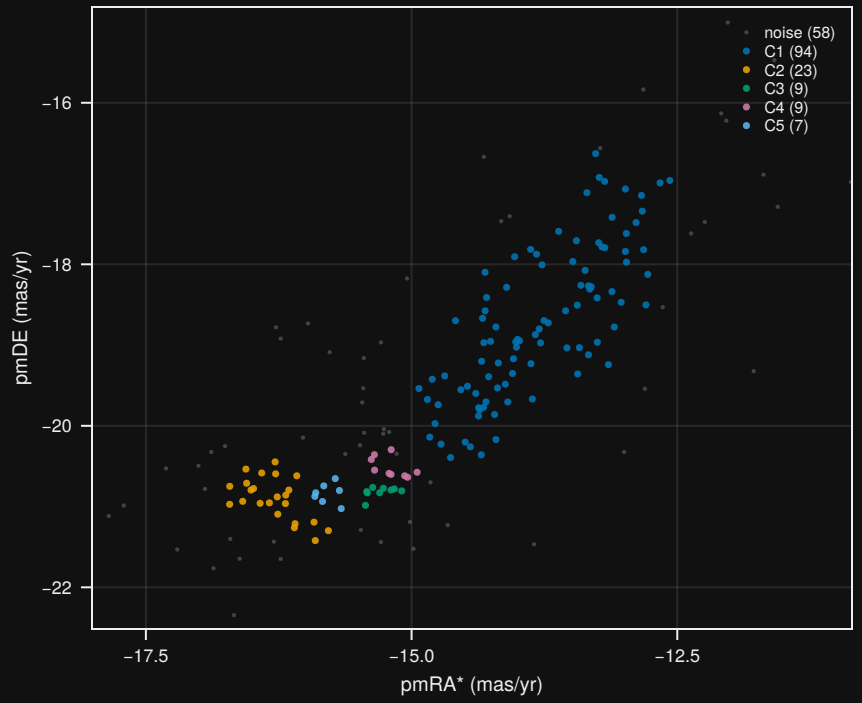


Parallax quality (n = 108)

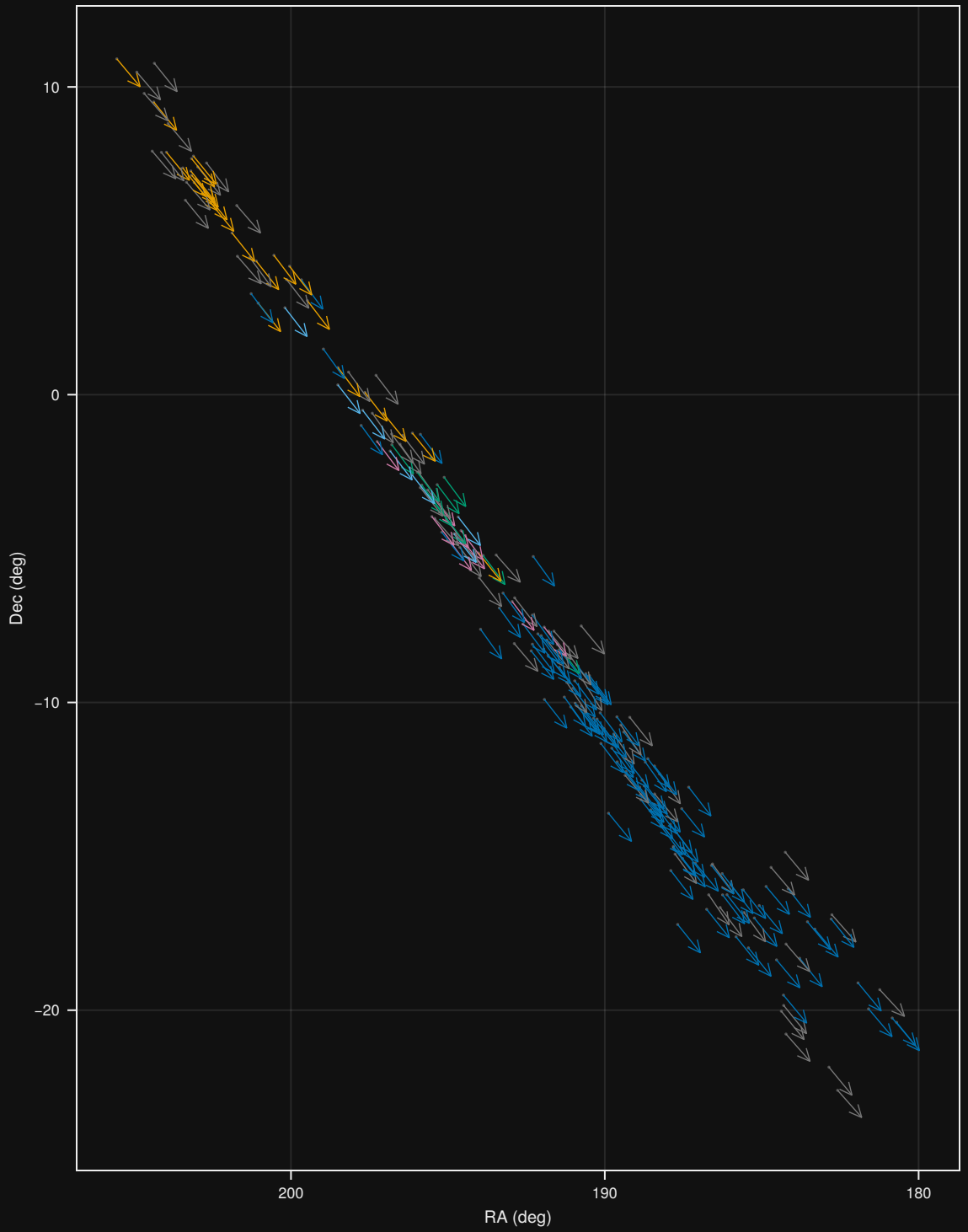


Gaia-1

PM clustering · HDBSCAN · 5 cluster(s) · noise 29%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	58	-15.28	-20.08	1.88	-1.63	6
C1	94	-13.85	-18.79	0.75	-1.46	14
C2	23	-16.28	-20.88	0.27	-1.55	3
C3	9	-15.30	-20.81	0.17	-1.65	4
C4	9	-15.19	-20.57	0.23	-2.94	0
C5	7	-15.83	-20.83	0.12	-1.21	1

Gaia-1

Per-cluster proper-motion vector maps

